Human-In-The-Loop Machine Learning (HITL ML)
Active Learning

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Outline

Introduction to HITL ML

Agenda

- Annotating unlabeled data
- Sampling unlabeled data items (active learning)
- Incorporating human–computer interaction into annotation
- Implementing transfer learning

Why HITL ML?

- $\sim$ 90% of ML applications are powered by **supervised learning**.
- Supervised models require **labeled data**, often created by humans.
- Humans (often non-coders) provide these examples (training data).
- Today's AI relies heavily on **intensive human feedback.**

Why HITL ML? Use Cases

- **Autonomous Vehicles:** thousands of hours of manual annotation (e.g., pedestrians, traffic signs).
- **Machine Translation:** models trained on human-translated texts for linguistic quality.
- **Interactive AI Systems:** systems learn more from non-technical users than from programmers.

How should humans & ML algorithms interact?

Defining HITL ML

Definition

A set of **strategies for combining human and machine intelligence.**

Objectives

- Increase ML model accuracy.
- Reach target/acceptable accuracy faster.
- Combine human/machine intelligence to maximize accuracy (synergy).
- Assist human tasks with ML to increase efficiency.

Components of HITL ML

- **Annotation**
  - Process of enriching raw data with tags (human knowledge).
  - **Purpose:** translates expert knowledge into usable training examples for ML models.
  - **Value:** the foundation of model performance, garbage in, garbage out.
- **Active Learning**
  - A strategy where the model selectively queries humans to label the most informative data points.
  - **Purpose:** maximize learning while minimizing labeling cost and time.
  - **Value:** efficient use of human effort, focusing on what matters most.

Components of HITL ML (cont'd)

- **Transfer Learning**
  - Leveraging pre-trained models and adapting them to new, domain-specific tasks.
  - **Purpose:** reduce training time and labeled data needs, especially in data-scarce environments.
  - **Value:** get a head start and avoid the cold-start problem.

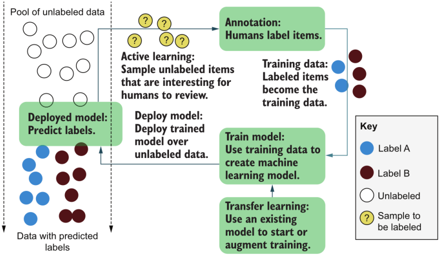
The HITL ML Process Cycle (Iterative Loop)

Start:

- **Unlabeled Data Pool**
  - Large collection of raw, unlabeled examples.

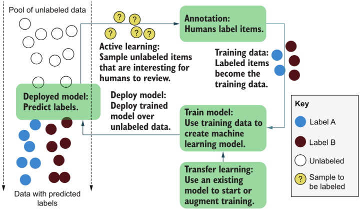
Optional Kick-Start:

- **Transfer Learning**
  - Use pre-trained models to initialize learning or improve sample selection.



The HITL ML Process Cycle (Iterative Loop)

- **Active Learning:** select the most "informative" or uncertain examples from the unlabeled pool.
- **Human Annotation:** domain experts label selected data.
- **Training Dataset Creation:** build or augment the labeled dataset.
- **Model Training:** train or fine-tune the model on the updated labeled data.
- **Model Deployment:** apply the trained model to new, unseen data.
- **Prediction:** generate labels for new data instances.



Annotation

Annotation

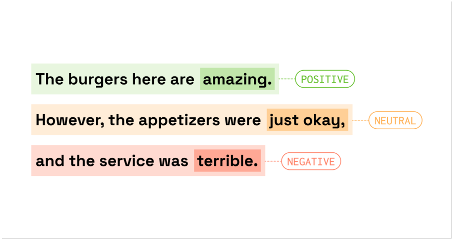
- **Annotation** is the process of **labeling raw data** (images, text, audio, video, etc.) to transform it into structured training data.
- It gives **meaning** to raw data, enabling models to learn patterns.
- High-quality annotations are **critical** for model performance.
- It defines what the model will learn.
- Often the **most labor-intensive phase** in the ML pipeline.
- Requires **precision, consistency**, and sometimes iterative revisions.
- May involve **millions of labeled examples** in large-scale projects.

Spectrum of Annotation Complexity: Simple Tasks

Simple Annotation Tasks

Tasks that can be done quickly with basic tools and minimal training.

- Sentiment labeling using HTML forms or spreadsheets (e.g., tagging a product review as positive, negative, or neutral).
- **Tools:** web forms, Google Sheets, Label Studio.
- **Characteristics:** high throughput; suitable for crowdsourcing (e.g., Mechanical Turk); low barrier to entry.

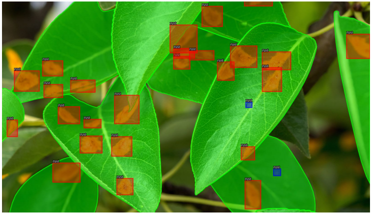


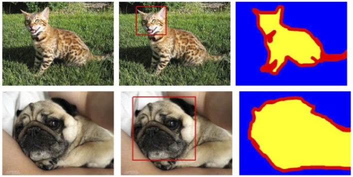
Spectrum of Annotation Complexity: Complex Tasks

Complex Annotation Tasks

Tasks requiring advanced tools, domain expertise, and significant preparation.

- Drawing **bounding boxes** around moving objects in video streams using specialized graphical interfaces.
- **Tools:** CVAT, Labelbox, custom-built annotation platforms.
- **Challenges:** annotation fatigue, human error, and the need for custom engineering (UI, storage, QA pipelines).
- **Effort:** months of development to create efficient workflows; involves UI/UX, tool optimization, and QA automation.





Data-Centric AI Shift & Academia–Industry Gap

- Performance gains often come from **improving data quality**, not just tweaking models.

Academic ML:

- Fixed datasets (e.g., ImageNet).
- Focus on algorithmic innovation.
- Ignores real-world data challenges.

Industry Reality:

- Data is noisy, incomplete, ever-changing.
- Success $=$ continuous data refinement.
- Co-optimization of models $+$ annotation strategies.

In modern ML, your data is as important as, if not more than, your model.

Why High-Quality Annotation is Hard

Human Error:

- Critical in high-stakes applications (e.g., autonomous vehicles).
- Errors are not random noise, they introduce **systematic bias**.
- "No algorithm survives truly bad training data."

Label Aggregation Challenges:

- Binary/objective tasks (e.g., spam detection) are easier to aggregate.
- Subjective/continuous tasks (e.g., bounding boxes, severity ratings) are more nuanced.
- ML tools can assist in reconciling disagreements.

Workforce Considerations:

- Quality depends on **finding, teaching, and managing** skilled annotators.
- Example: training annotators to do complex medical annotation.

"Better data beats more data, and humans are essential to making it better."

Active Learning

Active Learning

Definition

Active Learning is a machine learning approach where the model **selectively chooses which data samples should be labeled** by humans, rather than labeling everything.

Goal:

- Maximize model performance with **minimal labeled data.**
- Reduce **cost, time**, and annotation effort.

Why?

- More labeled data generally improves model performance.
- But human annotation is **slow, costly**, and error-prone.
- Active learning helps focus effort on the **most informative examples**.

Active Learning: Three Strategies

Uncertainty Sampling (Exploitation)

- Identify items near the **decision boundary** (model is “confused”).
- Likely to be misclassified, so labeling them moves the boundary.
- **Risk:** might focus on one part of the decision boundary.

Diversity Sampling (Exploration)

- Identify **underrepresented** or unknown items.
- Targets new, unusual, or underrepresented items for a complete picture.
- **Risk:** might focus on outliers far from the boundary, having little impact.

Random Sampling

- Seems simple but tricky (prefiltered data, changing data).
- Essential for gauging model accuracy and as a baseline.

Combining strategies is often the best approach to maximize both uncertainty and diversity.

Active Learning: Iterative Visualization

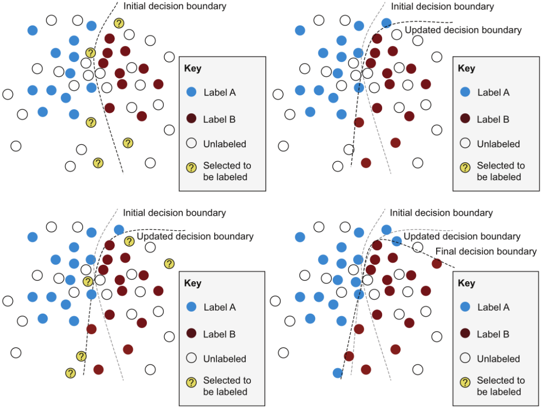


Figure: The iterative process of **active learning**: selected samples (yellow ?) update the decision boundary.

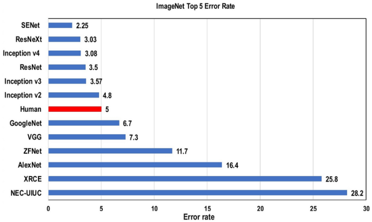
Active Learning: Defining “Random” for Evaluation Data

Random $\neq$ Representative

- Truly random sampling is difficult in practice:
  - Prefiltering bias.
  - Data drift over time or source.
- Real-world data often diverges from training sets.

ImageNet Case Study:

- Models achieve high accuracy on the benchmark (ImageNet).
- But fail on real-world images (e.g., social media posts).
- Shows the risk of overfitting to static, curated datasets.



Defining “Random” for Evaluation: A Better Approach

A Better Approach:

- Build **representative evaluation sets**:
  - Stratify by label distribution, time, and data source.
- Create **multiple test sets**:
  - In-domain (e.g., curated news).
  - Out-of-domain (e.g., noisy social media, low-resource regions).

When to Use Active Learning

Use Active Learning When:

- You can only annotate a **small fraction** of your total dataset.
  - Full labeling is infeasible (cost/time/resource constraints).
- Random sampling fails to capture data diversity.
  - Leads to underrepresentation of rare or edge cases.
- Annotation costs are extremely high.
  - Example: **video annotation**, 36,000 bounding boxes per minute of video; U.S. driving: $\sim$60 trillion hours/year; manually labeling even a fraction is unsustainable.
- You expect **data drift** or edge cases.
  - Active learning helps models adapt to evolving environments.

When *Not* to Use Active Learning

Don't Use Active Learning When:

- Critical applications require **full annotation.**
  - E.g., regulated medical records, financial documents.
- You operate in a **highly stable, static** environment.
  - Model already performs well and no new data domains are expected.
- The **cost of new data exceeds the value** of model improvement.
  - Important in scenarios with small performance gains or fixed ROI.

Machine Learning and Human-Computer Interaction (HCI)

Machine Learning and Human-Computer Interaction (HCI)

Machine Translation

- Initial efforts to improve human translation with machine translation faced **significant hurdles.**
- Simply having humans correct machine translation output proved **inefficient and prone to error perpetuation.**
- The core challenge wasn't just the accuracy of machine translation algorithms.

The Breakthrough: A User Interface Innovation

- The real solution emerged from advancements in the **user interface.**
- Modern translation systems adopted **predictive text**, similar to smartphones and email.
- Translators type naturally, accepting correct machine-generated word predictions with a simple key press.
- This significantly increased translation speed when predictions were accurate.

The most significant breakthrough was in human-computer interaction (HCI), not the underlying machine learning algorithm itself.

Effective integration of human input became the crucial factor.

User Interfaces: How Do You Create Training Data?

Why They Work:

- Leverage familiar **HTML form conventions**.
- Follow well-established **Human-Computer Interaction (HCI)** principles.
- Require **no user training**, users intuitively understand how to interact.

Design Matters:

- Consistency with standard form layouts ensures **efficiency and reliability**.
- Even small deviations (e.g., dynamic helper text) can cause confusion.
- Annotation tools should **minimize friction**, not add features that surprise users.

The Power of Simplicity: Binary Responses

Binary decisions (yes/no, correct/incorrect)

- Simple is often the best for quality control.
- Breaking down complex tasks into **binary decisions** simplifies interface design.
- This also facilitates the implementation of quality control measures.

Complex Interfaces, Complex Expectations

- More complicated annotation tasks (e.g., polygon annotation) $\Rightarrow$ more complex interface requirements.
- Users bring expectations from familiar software (e.g., Photoshop for image editing).
- Meeting these expectations can be development-intensive.

Mouse vs. Keyboard for Annotation

The Inefficiency of Mouse Usage

- For repetitive annotation tasks, mouse movement is **inefficient**.
- Aim to make the entire annotation process **keyboard-driven**.
- This includes annotation itself, form submission, and navigation.
- Keyboard-centric workflows can significantly improve annotator rhythm.

When a Mouse is Necessary: Rich Annotations

- If mouse usage is unavoidable, ensure the resulting annotations are **rich and valuable**.
- This justifies the slower input method.

Human Attention & Priming in Annotation

Why Attention Matters

- Human annotator focus and attention span are crucial for accurate training data.
- **Contextual effects** can alter perception and introduce errors.
- Linguistics research demonstrates that the presence of a *kiwi* or *kangaroo* influenced accent perception.

Priming: How Sequence Shapes Perception

- **Priming** describes how context or sequence influences human perception.
- A sequence of similar tasks can bias annotator responses.
- For example, annotating 99 negative-sentiment posts may lead to errors on the 100th post.

Diversify examples to keep the annotator focused.

Basic Principles for Designing Annotation Interfaces

Favor Binary Choices

- Simplifies decision-making and UI design.
- Enables easier quality control and consistency.
- Break complex tasks into a series of yes/no.

Diversify Task Sequences

- Prevents priming effects and cognitive bias.
- Mix samples to keep annotators engaged and alert.
- Randomize order to reduce repetition-induced errors.

Follow Established Interaction Conventions

- Build on standard HTML forms and familiar UI elements.
- Minimize need for training or onboarding.
- Avoid “clever” design that violates user expectations.

Support Keyboard Shortcuts

- Improves speed, ergonomics, and flow for power users.
- Reduces cognitive load compared to mouse-only interfaces.

Machine-Learning-Assisted Humans VS. Human-Assisted Machine Learning

Improve ML Models with Human Input

- Humans provide **labels, corrections**, or feedback.
- Helps models learn faster, generalize better, and reduce bias.

Examples

- Labeling rare edge cases.
- Reviewing model predictions in active learning.
- Enhancing LLMs with Human Feedback (**RLHF**).

Enhance Human Tasks with ML Assistance

- ML helps humans **make better decisions**.
- Tasks become faster, more scalable, and less error-prone.

Examples

- Auto-suggest in labeling tools.
- Pre-annotations with confidence scores.
- Autocomplete in message writing.

Search Engines: A Combined Approach

Human-Assisted Machine Learning

Users train the system

- Every click, scroll, or dwell time is **feedback**.
- Clicking the 4th result over the 1st? $\rightarrow$ signals "This result is more relevant."
- Patterns of behavior become **training data**.
- Search ranking adapts based on user interactions.
- No need for explicit annotation, behavior becomes feedback.

Machine-Learning-Assisted Humans

The system supports the user

- Search engines use ML to **rank, filter, and personalize** results.
- Algorithms trained on past behavior predict what's most useful.
- Users get faster, more accurate answers with less effort.
- The system evolves to better serve human intent over time.

Transfer Learning

Transfer Learning to Kick-start Your Models

Use What Already Exists

- Existing datasets often align closely with your goals.
- Example: adapt a product-review sentiment model for **movie reviews.**
- No need to collect labels from zero.

What is Transfer Learning?

- Reusing a model trained on one task to improve performance on a **related task.**
- Saves time, data, and compute.

Examples

- Use general pre-trained models (e.g., ImageNet, BERT, CLIP, GPT).
- **Fine-tune** them for your specific domain or use case.

Most modern ML projects succeed by adapting, not reinventing.

Transfer Learning: Neural Network View

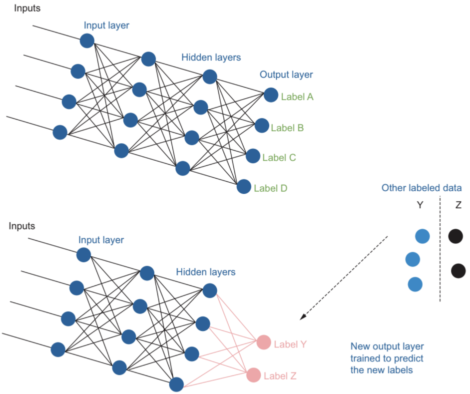


Figure: In **transfer learning**, early layers of a pre-trained network are reused; the output layer is replaced and retrained.

Transfer Learning in Computer Vision

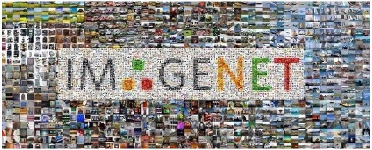
- Transfer learning has **significantly advanced** computer vision performance.

ImageNet

- Common strategy: **pre-train on ImageNet**.
- Millions of images across 1,000 diverse categories.

Reusing Learned Features

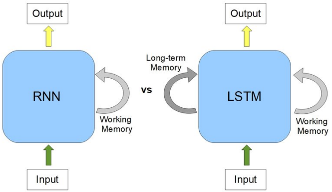
- Models learn general visual features: edges, textures, shapes.
- These features are **transferable** to other tasks (e.g., medical imaging, object detection).



Re-training the last layers enables accurate classification on new tasks, even with limited data.

Transfer Learning in NLP: Early Struggles

- **LSTMs & RNNs** dominated early NLP.
- **Problems:**
  - Poor generalization across tasks.
  - Struggled with long-range dependencies.
  - Required large amounts of task-specific labeled data.
- Progress in **text transfer learning** was slow compared to vision.



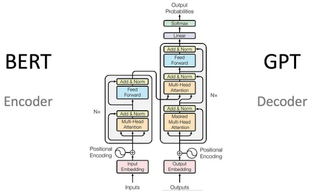
Transfer Learning in NLP: Transformer and LLM Breakthrough

Transformers (Vaswani et al., 2017):

- Used **self-attention** for better context handling.
- Enabled **parallel training** over large corpora.

BERT, GPT:

- Pre-trained on massive text (Wikipedia, web, books).
- Learned **general-purpose** language representations.
- Marked a shift: **pre-train once, fine-tune many tasks**.



Transfer learning & LLMs significantly improved efficiency and scalability in NLP applications.

Pre-training and Fine-tuning of LLMs

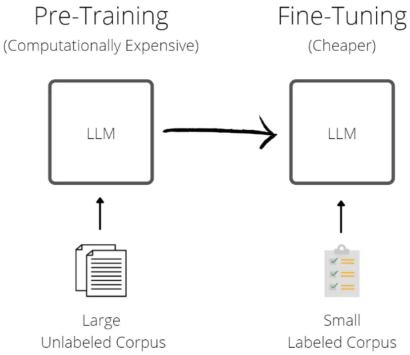


Figure: LLMs are first **pre-trained** on a large unlabeled corpus, then **fine-tuned** on a small labeled corpus for a specific task.

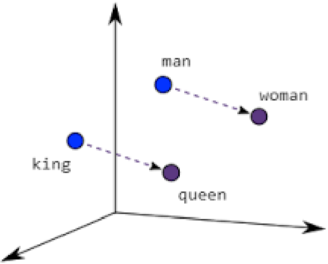
Learning Word Meaning from Context

> *"You shall know a word by the company it keeps."*
> , Firth, 1957

- Words used in similar contexts tend to have similar meanings.
- Example: *doctor* and *surgeon* often appear in medical contexts.

How Models Learn Meaning

- Represent words as **vectors** capturing contextual patterns.
- Similar contexts → closer vectors.



Free Supervision via Context Prediction

- **Self-supervised learning**: predict missing words from surrounding context.
- Example: *"My ___ is cute. He ___ playing."* $\Rightarrow$ *"dog", "is"*
- Enables training on massive unlabeled corpora, powering transfer learning.

Context prediction is the engine behind LLMs' ability to learn and transfer word meaning across tasks.

Knowledge Quadrant for Machine Learning

Knowledge Quadrant for Machine Learning

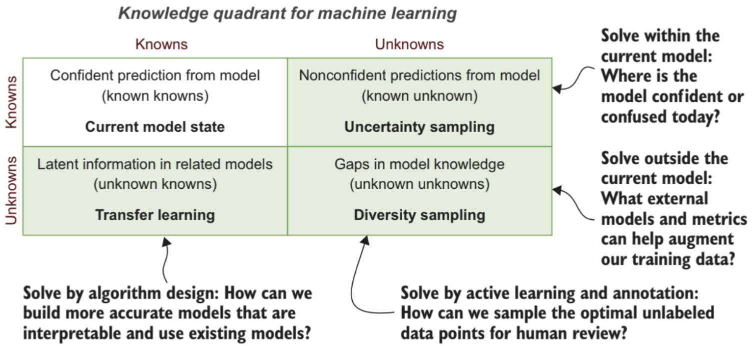


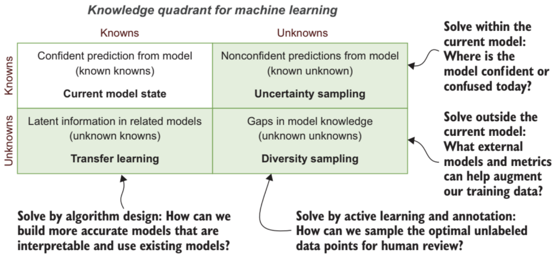
Figure: The **knowledge quadrant** maps what your model knows or does not know against what is inside or outside the model.

The Four Quadrants

- **Known knowns**, what your machine learning model can **confidently and accurately** do today. This quadrant is your model in its current state.
- **Known unknowns**, what your machine learning model **cannot confidently** do today. You can apply **uncertainty sampling** to these items.
- **Unknown knowns**, knowledge **within pre-trained models** that can be adapted to your task. **Transfer learning** allows you to use this knowledge.
- **Unknown unknowns**, gaps in your machine learning model. You can apply **diversity sampling** to these items.

Reading the Quadrant

- The **top row** captures your model's knowledge.
- The **bottom row** captures knowledge **outside** your model.
- The **left column** can be addressed by the **right algorithms**.
- The **right column** can be addressed by **human interaction**.



Summary

Key Takeaways

- **HITL ML is an Iterative Process**
  - Combines human judgment with machine learning to continuously improve models.
- **Annotation Techniques Are Critical in ML**
  - Essential for creating high-quality training data efficiently and accurately.
- **Human–Computer Interaction (HCI) Enhances Annotation**
  - Enables the design of effective, user-friendly annotation tools.
- **Active Learning Optimizes Label Acquisition**
  - **Uncertainty sampling**: targets data the model is least confident about.
  - **Diversity sampling**: selects a broad range of examples to maximize learning.
- **Transfer Learning Accelerates ML Applications**
  - Leverages pre-trained models to reduce annotation effort and improve accuracy.

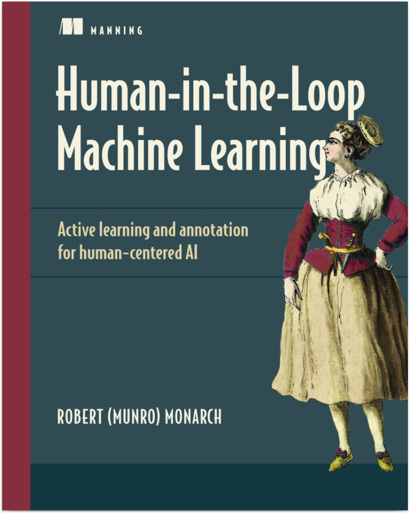
Outline

Reference Book

Robert (Munro) Monarch

Human-in-the-Loop Machine Learning: Active learning and annotation for human-centered AI.

Manning Publications.



Slide Acknowledgment

The slides used in this course are based on the official textbook materials, with modifications made where necessary to suit the course requirements and enhance the learning experience.